

# TrendChecker Presentation Guide

Speaker Script & Flow Structure | Prepared by Zarar

*Use this script/structure to present the project to your supervisor. It naturally flows from the basic "what it does" to the moderate technical "how it works," without getting bogged down in extreme complexity.*

## 1. Introduction: The Big Picture

**Basic** TrendChecker is a 100% local, multi-modal authenticity system. Its goal is to stop the spread of fake news, deepfakes, and AI-generated hallucinations on social media.

**Technical** Instead of relying on expensive cloud APIs, the system runs locally on standard hardware. It processes Text, Claims, Images, and Video simultaneously using a composite scoring engine.

## 2. Stage 1: The Capture Pipeline

**Basic** We start by ingesting data. The system scrapes live posts from platforms like X (Twitter), Reddit, Facebook, YouTube Shorts, and Instagram Reels.

**Technical** The scrapers pull down the raw text, metadata, and media files directly into local RAM for immediate processing.

## 3. Stage 2: Text Analysis (Is it AI-written?)

**Basic** First, we check if the caption or post was written by a human or a chatbot like ChatGPT.

**Technical** We use a model based on Lexical Shannon Entropy.

- AI models choose the most mathematically probable words, which results in very predictable, low-entropy text.

- Human writing is chaotic and uses diverse vocabulary (high entropy). By measuring this entropy, we can accurately flag AI-generated text.

#### 4. Stage 3: Claim Verification (Is it true?)

**Basic** Next, we fact-check the actual claims made in the post. If someone posts "The Eiffel Tower burned down," we need to verify it.

**Technical** We use a Domain Exclusion & Consensus algorithm.

- **The old flaw:** Systems used to search the web, find the exact same viral post on another site, and use it as proof.
- **Our fix:** We extract the entities, search the web, but block the original domain. We require a Multi-Source Consensus—meaning 3 to 5 highly trusted, independent news sources (like Reuters or AP) must confirm the claim for it to be marked "Supported."

#### 5. Stage 4: Image Forensics (Are the photos real?)

**Basic** We check if the attached images are real camera photos or AI-generated by tools like Midjourney or DALL-E. We don't look for hidden metadata (because social media strips that away); we look at the actual pixels.

**Technical** We use 2D Fast Fourier Transform (FFT) and PRNU Noise Kurtosis.

- Real camera sensors have physical imperfections that leave a unique, invisible layer of "noise" (PRNU) on every photo.
- AI generators don't have physical lenses. They generate images that are mathematically too smooth. We convert the image into frequency waves (FFT) and look for artificial spikes (Kurtosis). Natural noise passes; artificial spikes fail.

#### 6. Stage 5: Video Forensics (Are the videos deepfakes?)

**Basic** For YouTube Shorts and IG Reels, we check for AI video generation (like OpenAI Sora) or deepfake manipulation.

**Technical**

We use the Gunnar Farneback Optical Flow algorithm.

- We extract keyframes and calculate "motion vectors"—how pixels move from frame 1 to frame 2.
- Real videos obey physics (consistent motion blur). AI videos hallucinate frames, causing pixels to "morph" or teleport unnaturally. The Optical Flow algorithm detects these impossible physical movements.

## 7. Conclusion: The Authenticity Engine

**Basic**

Finally, we don't just rely on one check. We combine all of them.

**Technical**

The Multi-Signal Composite Engine takes the entropy score, the consensus count, the image kurtosis, and the optical flow vectors to generate one final Confidence Score (0–100%).

*Closing Statement for Supervisor: "By combining lexical analysis, independent consensus, and pixel-level motion tracking entirely on a local machine, TrendChecker offers a highly accurate, privacy-first defense against modern misinformation."*